

IMO Training – Number Theory

Law of Quadratic Reciprocity

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Introduction

Unless otherwise specified, we shall focus on integers. Furthermore, let p be an odd prime.

Definition. Let $(a, m) = 1$. Then a is a *quadratic residue mod m* if $\exists x \in \mathbb{Z}$ such that $x^2 \equiv a \pmod{m}$. Otherwise, we call a is a *quadratic nonresidue mod m* .

Definition. The *Legendre's Symbol* $\left(\frac{a}{p}\right)$ is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} +1 & \text{if } a \text{ is a quadratic residue mod } p \\ -1 & \text{if } a \text{ is a quadratic nonresidue mod } p \\ 0 & \text{if } p \mid a \end{cases}$$

Proposition. (Euler's Criterion) For all a , $\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$.

Proposition. $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right)$

Example. Find $\left(\frac{-1}{p}\right)$ and $\left(\frac{2}{p}\right)$.

Theorem. (Quadratic Reciprocity Law) Let p, q are distinct odd primes, then

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\left(\frac{p-1}{2}\right)\left(\frac{q-1}{2}\right)}.$$

Example. Evaluate the following:

Example. Prove that -3 is not a primitive root of any prime of the form $4^n + 3$.

Example. Solve $x^2 + 3xy - 2y^2 = 122$ for integers.

Example. Show that there are infinitely many primes of the form $4k + 1$.

Example. Find all integers a, b s.t. $2b^2 + 3 \mid a^2 - 2$.

Exercises

1. (Bulgaria 1998) Let $m, n \in \mathbb{N}$ such that $A = \frac{(m+3)^n + 1}{3m} \in \mathbb{Z}$. Prove that A is odd.
2. (IMO 1996) Let $a, b \in \mathbb{N}$ s.t. $15a + 16b, 16a - 15b$ are both perfect squares. Find the minimum value of the smaller square.
3. (Bulgaria 1996) Find all primes p, q such that $pq \mid (5^p - 2^p)(5^q - 2^q)$.
4. (USA TST 2003) Find all ordered primes p, q, r s.t. $p \mid q^r + 1, q \mid r^p + 1, r \mid p^q + 1$.
5. (China 2009) Find all primes p, q s.t. $pq \mid 5^p + 5^q$.

Extra Space